Value-path attribution: one fixed target, two spatial readouts

a Fix the internal target



Clean CLS attention A(x)[0, j]



Direction u

$$g(z) = \sum_{j=0}^{196} A(x)_{0j} V(z)_j$$

$$u = \frac{g(x) - g(x_0)}{\|g(x) - g(x_0)\|_2}$$

$$f(z) = u^\top g(z)$$

A and u stay fixed along the path.

Value features vary with the input.

b Integrate along the input path



x_0 x_α x

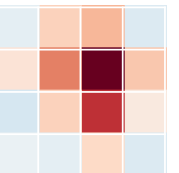
$$x_\alpha = x_0 + \alpha(x - x_0)$$

$$R_i = (x_i - x_{0i}) \int_0^1 \partial_i f(x_\alpha)\, d\alpha$$

Signed input-space IG

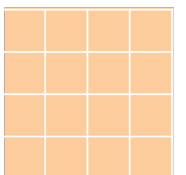
RGB channels are summed with sign.

c Resolve spatial contributions



NNS

Input-space contributions



Patch baseline

c_j/16 per fine cell

$$c_j = A(x)_{0j} u^\top [V(x)_j - V(x_0)_j]$$

196 patch contributions + separate CLS scalar

negative ← 0 → positive